João André **Roque Costa**

Software, Data & AI Engineer

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Computer Science and Engineering Master's graduate focused on **Artificial Intelligence**, **Data Science**, and practical data systems. At Deloitte, delivered an Azure migration involving **Databricks** and **Azure Data Factory**, built a **Python** reporting-automation prototype, and contributed to **RAG**, **OCR**, and **LLM** data-migration work. Supported by research in scalable nonlinear dimensionality reduction and self-hosted software projects, I bring a pragmatic foundation for turning complex data and AI ideas into reliable, useful systems while communicating technical decisions clearly.

Professional Experience

Tech Analyst — *Coimbra, Portugal*
Deloitte — *Apr. 2026 - Oct. 2026*

- Delivered a resource-group migration involving **Databricks** and **Azure Data Factory** assets to a new Azure subscription within the planned timeframe.
- Built a working **Python** and **Playwright** prototype to automate HR report generation.
- Contributed technical context to a client-facing proposal for documentation automation using **RAG** and **OCR**.
- Helped deliver a cross-functional **LLM** prototype for data migration using **Azure AI Foundry**.
- Worked with Azure Blob Storage, Azure Managed Instances, MongoDB, MySQL, and Git while ramping up on an existing data-focused project structure.

Teaching Assistant — *Lisbon, Portugal*
Instituto Superior Técnico — *Oct. 2024 - Jul. 2025*

- Lectured and evaluated practical classes, helping students apply **Python**, pandas, and matplotlib to data treatment, visualization, and analysis.
- Tracked student progress, assisted with practical projects, and adapted technical explanations and feedback to different audiences.

Education

M.Sc. Computer Science and Engineering — *Lisbon, Portugal*
Instituto Superior Técnico — *Sep 2023 - Nov 2025*

- M.Sc. in Computer Science and Engineering from Instituto Superior Técnico, with a self-directed focus on **Artificial Intelligence** and **Data Science**; final grade 15/20.
- Relevant Coursework: Deep Learning, Machine Learning, Data Science, Data Analysis and Integration, Cloud Computing and Virtualization, Autonomous Agents and Multi-Agent Systems, Planning, Learning and Intelligent Decision Making, Optimization Algorithms
- Thesis: Maximum Variance Unfolding on Disjoint Manifolds (MVU-DM)
  - Developed a decomposition-based extension to **MVU** for nonlinear dimensionality reduction on disconnected datasets.
  - Designed a parallelizable approach for large-scale datasets while preserving local geometric structure in comparable embeddings.

B.Sc. Computer Science and Engineering — *Lisbon, Portugal*
Instituto Superior Técnico — *Sep 2020 - Jul 2023*

- Relevant Areas: **Algorithms**, distributed systems, operating systems, database theory, systems thinking, C/C++, SQL
  - Built foundations in **distributed systems**, including scalability, fault tolerance, security, remote calls, and distributed naming.
  - Applied relational schema design, implementation, and transaction management in database coursework.

Extracurricular

- Built **Autolab**, a self-hosted modular system on a Linux NUC with per-machine resource management, product price tracking, and a virtual dashboard for monitoring.
- Developed a Discord-based matchmaking application integrating the **Discord API**, PostgreSQL, and Redis for ELO ranking, match history, caching, and queue management.
- Built price-tracking and marketplace automation tools using **Python**, Selenium, BeautifulSoup, web interfaces, and multi-platform API integration.

Skills & Competencies

AI and Machine Learning	**Python**, TensorFlow, Keras, scikit-learn, Deep Learning, Machine Learning, LLM-based automation, RAG, OCR, nonlinear dimensionality reduction
Cloud and Data Platforms	**Azure**, Databricks, Azure Data Factory, Azure AI Foundry, Azure Blob Storage, Azure Managed Instances, MongoDB, MySQL
Data and Analytics	**SQL**, PostgreSQL, NumPy, pandas, matplotlib, data analysis, reporting automation, optimization
Software and Automation	**Playwright**, Selenium, BeautifulSoup, Docker, Linux, Git, RESTful APIs, Discord API
Communication and Delivery	technical writing, stakeholder communication, cross-functional collaboration, practical-class lecturing, mentoring, structured problem-solving